Native development checks — 29 October 2026

These are local development tests, not the requested real-curve experiment. No performance isolation or clock is claimed. The tested source and command bindings are in BUILD.json; all retained attempts belong to the local review bundle named in the delivery record.

The final build-controls.sh invocation compiled the C++17 PARI producer, the separate GMP verifier and the tiny fixture generator under -O2 -Wall -Wextra -Werror. Both native self-tests, one complete synthetic integration and nine corrupted-evidence controls passed. The fixture repeats six tiny fields into 21 rows; its 1,127 inherited features are dummy values. The verifier checked 588 point rows/100,548 trait cells, 19,152 cutoffs, 119,700 point differences and 28,175 dummy-feature differences. All eleven unknown historical-point stand-ins remained NA. Both programs share OpenSSL for SHA; their group and trait arithmetic is independently implemented.

The nine rejection controls alter a point, a trait, an unknown generator, a recipe flag, a cutoff key, a difference, a required point row, an inherited matrix cell and the headline summary. Each exits 1 for its intended diagnostic and writes no admission receipt. Their data come only from the synthetic fixture.

Retained producer attempts 001–002 failed compilation; 003 compiled but its tiny doubling control failed because PARI modular coordinates had not been lifted to integers. Attempt 004 fixed that defect with liftall; 005–007 added identity/bit-width, synthetic classification, Hasse and transport-label checks. Verifier 001–002 failed strict indentation diagnostics; 003–005 passed the progressively strengthened synthetic checks. The final combined build also independently recomputes and compares the headline summary; an added ninth control rejects a false summary. The build uses the current sources, not a previously compiled binary.

The five conditional Lean proofs compiled. Axiom-auditor attempts 001 and 003 hit their 1 GiB/2 GiB limits; 002 lacked lean on the child process PATH. Attempt 004 used an explicit PATH and 4 GiB cap and passed: six declarations including the generated equation theorem, zero failing, zero compiler-trusted. Only propext and Quot.sound occur among the reported theorem foundations. There is no lean4checker fresh recheck or whole-repository gate claim.

An initial inherited-input copy had 1,100 features rather than the required 1,127. It was retained outside the repository and replaced with all 21 hash-matched final-replay files before freezing inputs or any real-cohort run. No production cohort was executed with those stale inputs or the defective producer. The diagram separates development from the blocked actual comparison. No attack, historical criterion or speed result follows.

